

MSS (2004) Revisited:

Weak Instruments and Panel-Internal Identification

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KU Leuven: D0S91a Advanced Applied Econometrics

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May 2026

Abstract

We re-examine the rainfall-instrumental-variables design of Miguel et al. (2004) using post-2005 weak-instrument diagnostics and alternative panel identification strategies taught in our course. On the original panel of 41 sub-Saharan African countries (1981-1999), the rainfall first stage fails the Stock and Yogo (2005) 10%-maximum-size critical value by a factor of seven; the 2SLS point estimate carries the wrong sign, the Anderson-Rubin 95% confidence set is $[-4.41, 7.49]$, and a cluster-bootstrap percentile interval covers the entire interesting parameter space. Three panel-internal identification strategies that do not rely on rainfall: first-difference IV (Anderson and Hsiao, 1981) and difference- and system-GMM (Arellano and Bond, 1991; Blundell and Bond, 1998), all recover the negative MSS sign, with the difference-GMM coefficient -0.29 significant at the 10% level. The data inform the sign of the income-conflict relationship across distinct identifying assumptions, but pin down its magnitude only loosely under any single design available within the original panel.

1 Introduction

Estimating the causal effect of income shocks on civil conflict is not an estimation problem but an identification problem. OLS confounds three biases: conflict destroys output (reverse causation), national accounts in low-income economies are measured with error, and persistent country traits affect both (omitted variables). No amount of data fixes this. The effect is identified only if we can find a source of variation in growth that is exogenous to conflict and does not enter the conflict equation directly.

Miguel et al. (2004), henceforth MSS, proposed that source: annual rainfall growth, used as an excluded instrument for per-capita GDP growth on a panel of 41 sub-Saharan African countries over 1981-1999. The identifying assumptions are that rainfall shifts agricultural output (relevance), and that it affects conflict only through that channel (exclusion). The paper was novel in 2004. It also predates several of the diagnostic tools we now use to evaluate IV designs.

The MSS design has since been challenged on three grounds: the relevance and functional form of the first stage (Ciccone, 2011), the credibility of an alternative price-based instrument (Brückner and Ciccone, 2010), and the exclusion restriction itself (Sarsons, 2015). Section 3 reviews these critiques.

We revisit MSS with two questions. First, how much the original rainfall design tells us about β once modern weak-instrument diagnostics and weak-IV-robust inference are applied to the 2SLS output. Second, whether panel-internal identification strategies that do not rely on rainfall, from Anderson and Hsiao (1981) first-difference IV to the difference- and system-GMM estimators of Arellano and Bond (1991) and Blundell and Bond (1998), recover a coherent estimate.

Section 2 formalises the MSS identification. Section 3 surveys the critiques. Section 4 describes the data. Section 5 sets out the estimators. Section 6 reports estimates. Section 7 concludes.

2 The MSS identification strategy

MSS fit the structural equation

$$C_{it} = \alpha_i + \lambda_i t + \beta g_{it}^y + \gamma' X_{it} + u_{it}, \quad (1)$$

where $C_{it} = \mathbf{1}\{\text{conflict}_{it}\}$ is a binary indicator for civil conflict in country i at year t , α_i are country fixed effects, $\lambda_i t$ are country-specific linear time trends, $g_{it}^y = \Delta \log(\text{GDP}_{it})$ is per-capita GDP growth, and X_{it} collects time-varying controls (lagged Polity 2). Opportunity-cost theory (Grossman, 1991) predicts $\beta < 0$.

Estimating equation (1) by OLS is inconsistent for the reasons listed in Section 1. MSS instrument g_{it}^y with annual rainfall growth g_{it}^R and its first lag:

$$g_{it}^y = a_i + b_i t + \pi_1 g_{it}^R + \pi_2 g_{i,t-1}^R + \delta' X_{it} + v_{it}. \quad (2)$$

The identifying assumptions, made conditional on α_i and X_{it} , are:

Relevance: $\text{Cov}(g_{it}^R, g_{it}^y) \neq 0$. Rainfall shifts agricultural output, which is a large share of African GDP. Testable through the first-stage F statistic on the excluded instruments.

Exclusion: $\text{Cov}(g_{it}^R, u_{it}) = 0$. Rainfall affects conflict only through GDP. Not directly testable in general; with two instruments, the Sargan test provides one degree of freedom of indirect check on whether the two rainfall measures imply the same β .

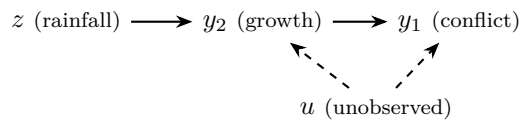


Figure 1: The MSS identification strategy as a DAG. Solid arrows: assumed causal pathway ($z \rightarrow y_2 \rightarrow y_1$). Dashed arrows: unobserved confounders that bias OLS. Identification of β requires the $z \rightarrow y_2$ arrow (relevance, testable) and the *absence* of any $z \rightarrow y_1$ arrow (exclusion, not directly testable).

The specification choices matter, which is why MSS use country fixed effects to absorb time-invariant heterogeneity (geography, institutions, colonial history) and country-specific linear time trends to absorb slow secular movements (democratisation, technology diffusion). They do not use year fixed effects, which would absorb the common component of rainfall shocks across

the panel and reduce the variation the instrument exploits. Standard errors are clustered at the country level.

Their preferred 2SLS specification yields $\hat{\beta} \approx -2.4$: a five-percentage-point decline in growth raises the probability of conflict by roughly twelve percentage points.

3 Methodological critiques and alternative identification

The literature that followed MSS extended the design along three axes: it offered alternative instruments, questioned the first-stage specification, and tested the exclusion restriction directly. Only the last two are critiques in the strict sense; the first is a parallel identification strategy.

Alternative instrument. Brückner and Ciccone (2010) replace rainfall with country-specific commodity export price shocks (a weighted sum of global commodity prices using each country’s export composition as weights), instrumenting growth in the same structural equation (1). The exogeneity argument is small-country price-taking: global commodity prices are determined on world markets, and no single sub-Saharan African country is large enough to move them. They recover a negative and significant β , corroborating the MSS sign under a different identifying assumption. This is identification pluralism rather than critique: two unrelated exogeneity arguments converge on the same conclusion, which strengthens MSS rather than weakens it. Within MSS’s own design, the analogue is the Sargan over-identification test on the two rainfall instruments (current and lagged); we return to it in Section 6.

Specification critique. Ciccone (2011) attacks the functional form rather than the exogeneity. He shows that the original MSS result stems from a counterintuitive *positive* correlation between conflict at t and rainfall levels at $t - 2$, rather than from negative rainfall shocks lowering income and raising conflict through the agricultural channel. Re-examining the analysis with updated data, he reports that conflict is largely unrelated to rainfall. Miguel and Satyanath (2011) reply arguing that the MSS findings remain when rainfall levels are used as instruments, while acknowledging that the first-stage relationship weakens after 2000. The lesson is generic: an exogenous instrument applied in the wrong functional form still produces an inconsistent estimate.

Direct test of exclusion. The most consequential critique tests the exclusion restriction itself. Sarsons (2015) uses Indian district-level data on rainfall and Hindu-Muslim religious riots, combined with records of irrigation dams. In districts *downstream of dams*, water supply is decoupled from local rainfall: income in these districts is much less sensitive to rainfall fluctuations, so the agricultural channel is mechanically attenuated. If MSS’s exclusion holds (rainfall affects conflict only through agricultural income), then in dam-downstream districts rainfall should not predict conflict. She finds that rain shocks remain equally strong predictors of riot incidence in downstream districts. The implication is a direct $z \rightarrow y_1$ arrow that the MSS DAG denies (Figure 2).

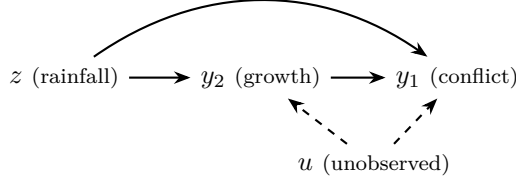


Figure 2: Sarsons’s critique as a DAG. Same as Figure 1 plus a direct $z \rightarrow y_1$ arrow that exclusion denies. Evidence from regions where the $z \rightarrow y_2$ channel is physically blocked (downstream of dams) supports the existence of this arrow.

The methodological lesson is general. Exclusion restrictions are sometimes partially testable: if the IV is claimed to operate through one channel, find a setting where the channel is shut and check whether the IV still moves the outcome. The post-2015 standard for IV papers in development economics has internalised this test.

4 Data

We use the replication panel distributed by MSS via the Harvard Dataverse (doi:10.7910/DVN/27324). It covers 41 sub-Saharan African countries over 1981-1999. After constructing one lag of GDP growth and dropping country-years with missing controls, the estimation sample is 702 country-year observations.

The outcome `any_prio` is a PRIO indicator equal to one if a country experiences at least 25 battle deaths in a year. The endogenous regressor `gdp_g` is annual per-capita GDP growth. The excluded instruments are current and lagged rainfall growth, `GPCP_g` and `GPCP_g_1`, constructed from the Global Precipitation Climatology Project. Controls follow MSS: initial log income, lagged Polity 2, ethnic and religious fractionalisation, an oil-exporter dummy, log mountainous terrain, and log lagged population. Only lagged Polity 2 is time-varying; the rest are absorbed by country fixed effects in our preferred specifications.

Table 1: Descriptive statistics, estimation sample ($N = 702$).

Variable	N	Mean	SD	Min	Max
<code>any_prio</code>	702	0.269	0.444	0.000	1.000
<code>gdp_g</code>	702	−0.006	0.066	−0.474	0.319
<code>GPCP_g</code>	702	0.015	0.210	−0.550	1.680
<code>y_0</code>	702	1.16	0.90	0.316	4.84
<code>polity21</code>	702	−3.55	5.55	−10	9
<code>ethfrac</code>	702	0.655	0.237	0.036	0.925
<code>relfrac</code>	702	0.487	0.185	0.000	0.783
<code>Oil</code>	702	0.117	0.321	0	1
<code>lmtnest</code>	702	1.58	1.43	0	4.42
<code>lpop11</code>	702	8.77	1.20	5.75	11.7

Three features of the panel matter for what follows. The unconditional conflict frequency is

27%. Average growth is slightly negative (-0.6% per annum). The panel is democracy-poor on average (mean Polity 2 of -3.5 on a scale of -10 to 10).

5 Methodology

The estimators below differ in how aggressively they purge the endogeneity sources of Section 1 from the structural equation. POLS uses all variation jointly and trusts that endogeneity bias is small. FE removes the time-invariant part of u . 2SLS exploits the rainfall instrument (the $z \rightarrow y_2$ arrow in Figure 1) and assumes no direct $z \rightarrow y_1$ arrow. The cluster bootstrap and the Anderson-Rubin test are inference tools, not identification strategies, and supplement 2SLS when conventional asymptotics are unreliable. First-difference IV with lagged levels and the dynamic-panel GMM estimators identify β without using rainfall at all, relying on internal panel variation instead.

5.1 Pooled OLS (LPM)

We begin by estimating equation (1) by OLS, $\hat{\beta}_{\text{POLS}} = (X'X)^{-1}X'C$, treating the binary outcome as a linear probability model. Consistency requires $\text{Cov}(g_{it}^y, u_{it}) = 0$, which the three endogeneity sources in Section 1 rule out. POLS therefore serves as a benchmark, not as a causal estimate. The LPM is heteroscedastic by construction since $\text{Var}(u | x) = p(x)(1 - p(x))$ depends on x ; we report the Breusch-Pagan and White tests as confirmation, and cluster standard errors at the country level throughout to handle within-country correlation.

5.2 Fixed and random effects

Fixed-effects estimation removes the time-invariant part of u . Let $u_{it} = c_i + e_{it}$ with c_i constant within country; the within transformation subtracts country means from each variable, giving an estimator that is consistent if $\text{Cov}(g_{it}^y, e_{it}) = 0$ after demeaning. We report two-way FE (country and year) as our baseline and add a specification with country-specific linear trends in place of year FE, the original MSS choice. The two differ in how they treat the common component of rainfall variation across the panel: year FE absorb it, country trends preserve it. Random effects is consistent under the stronger assumption $\text{Cov}(c_i, x_{it}) = 0$ and is more efficient when it holds; the Hausman test compares FE and RE. Time-invariant controls (initial income, fractionalisation, terrain, oil, lagged population) are absorbed by country FE and do not appear in the FE or RE specifications.

5.3 Two-stage least squares

We then implement the MSS strategy: instrument g_{it}^y with current and lagged rainfall growth from equation (2), obtain fitted values $\hat{g}_{it}^y$ from the first stage, and plug them into the structural equation. Two assumptions support consistency.

Relevance. We test it through the cluster-robust F statistic on the excluded instruments in the first stage (Staiger and Stock, 1997), compared to the Stock and Yogo (2005) 10%-maximum-size critical value of 19.93 (two IVs, one endogenous regressor). Values below the threshold indicate weak instruments and invalid conventional standard errors.

Exclusion. Not directly testable. With two instruments we report the Sargan over-identification test, which asks whether both rainfall measures imply the same β . Rejection indicates that at least one instrument is correlated with u . Failure to reject is weak evidence at best given the single degree of freedom and the fact that both instruments share the same exogeneity argument; the test cannot detect homogeneous invalidity.

We additionally report the Wu-Hausman endogeneity test, which compares OLS and 2SLS and rejects exogeneity of g_{it}^y if the two estimates differ by more than sampling noise. Failure to reject can equally well reflect low power: under a weak instrument the 2SLS estimate becomes so imprecise that the test cannot detect any systematic difference from OLS.

5.4 Cluster bootstrap and weak-IV-robust inference

When the first-stage F falls short of conventional critical values, the standard 2SLS t -statistic does not have an asymptotic standard normal distribution and reported confidence intervals can have actual coverage far below their nominal level. We supplement the conventional output with two robust procedures. The first is a non-parametric cluster bootstrap: countries are resampled with replacement, the 2SLS coefficient is re-estimated on each resampled panel, and the percentile interval is reported from $B = 500$ replications. Resampling at the country level preserves the within-country correlation structure that cluster-robust standard errors assume.

The second is the Anderson and Rubin (1949) test, which is the theoretically correct route under genuinely weak instruments. It tests $H_0: \beta = \beta_0$ by checking whether the instruments still have explanatory power in the reduced form once β_0 is imposed; the resulting confidence set has correct coverage under arbitrarily weak first stages. We invert the test on a fine grid to obtain the 95% confidence set for β .

5.5 First-difference IV with lagged levels

If rainfall is too weak to identify β externally, the panel itself contains a second source of variation. First-differencing equation (1) eliminates the country fixed effects:

$$\Delta C_{it} = \beta \Delta g_{it}^y + \gamma' \Delta X_{it} + \Delta u_{it}.$$

Anderson and Hsiao (1981) propose instrumenting the differenced regressor Δg_{it}^y with its lagged level $g_{i,t-1}^y$, under the assumption $\text{Cov}(g_{i,t-1}^y, \Delta u_{it}) = 0$: past growth is uncorrelated with the change in unobserved conflict shocks. This identifying assumption is logically separate from the rainfall exclusion restriction; the two strategies fail or succeed for different reasons. With a single instrument the design is exactly identified, so the Sargan test does not apply; the Wu-Hausman diagnostic from Section 5.3 still does.

5.6 Difference and system GMM

Conflict is highly persistent: the lagged outcome is a strong predictor of current conflict. Extending equation (1) with $C_{i,t-1}$ as a regressor turns it into a dynamic panel,

$$C_{it} = \alpha_i + \rho C_{i,t-1} + \beta g_{it}^y + \gamma' X_{it} + u_{it}. \quad (3)$$

Arellano and Bond (1991) first-difference equation (3) to eliminate α_i and instrument the differenced lagged outcome $\Delta C_{i,t-1}$ with deeper lags of the level outcome $C_{i,t-2}, C_{i,t-3}, \dots$; the endogenous Δg_{it}^y is instrumented similarly. Blundell and Bond (1998) extend this with additional moment conditions on the levels equation (system-GMM), more efficient when the series are persistent. We use two-step estimators with cluster-robust SE and collapse the instrument matrix to lags two and three to keep the instrument count well below $N = 41$ (Roodman, 2009). Validity is assessed via the Sargan-Hansen test on the over-identifying restrictions and the Arellano-Bond m_2 test for second-order autocorrelation in the differenced residuals; failure to reject m_2 is a necessary condition for the moment conditions to hold.

6 Results and diagnostics

Table 2 reports the seven static specifications described in Section 5. Table 3 reports the diagnostic battery and Table 4 the panel-internal identification strategies (Anderson-Hsiao, Arellano-Bond, Blundell-Bond).

Table 2: Static specifications. Cluster-robust SE at country level in parentheses. Models 2-6 use two-way FE; model 7 replaces year FE with country-specific linear time trends. Time-invariant controls are included in POLS only and absorbed by country FE elsewhere.

	POLS (LPM)	FE 2-way	RE 2-way	Red. form	First stage	2SLS	2SLS (trends)
GDP growth	-0.507** (0.222)	-0.460** (0.220)	-0.443** (0.217)			0.141 (1.330)	-1.297 (1.071)
GDP growth (lag)	0.020 (0.243)	-0.173 (0.221)	-0.121 (0.277)			-0.165 (0.229)	0.040 (0.185)
Rainfall growth				0.024 (0.055)	0.042** (0.018)		
Rainfall growth (lag)				-0.084* (0.048)	0.025 (0.016)		
Polity 2 (lag)	0.000 (0.004)	-0.004 (0.005)	-0.002 (0.005)	-0.003 (0.005)	-0.001* (0.001)	-0.003 (0.006)	-0.004 (0.005)
N	702	702	702	702	702	702	702
R^2	0.129	0.012	0.006	0.005	0.019	0.569	0.710

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

POLS reproduces the MSS direction: a one-standard-deviation drop in growth (≈ 6.6 percentage points) raises the conflict probability by $0.51 \times 0.066 \approx 3.4$ percentage points, significant at the 5% level. FE and RE shrink the coefficient slightly towards zero ($\hat{\beta} = -0.46, -0.44$) with widened standard errors, as expected when between-country variation is stripped out. The reduced form shows that rainfall growth enters the conflict equation only weakly (lagged rainfall has $p = 0.08$). The first stage confirms that current-year rainfall growth has a statistically significant but quantitatively modest effect on GDP growth.

The 2SLS estimate (column 6) is the striking column. The point estimate is +0.14 with a cluster-robust SE of 1.33: the wrong sign, no statistical significance, and a confidence interval

covering essentially all economically relevant magnitudes. This collapse of the IV estimate relative to the well-determined POLS estimate is the canonical fingerprint of a weak instrument. Column 7 re-estimates the same model with country FE plus country-specific linear time trends (the MSS specification) and recovers $\hat{\beta} = -1.30$ (SE 1.07, $p = 0.23$): the correct sign, larger in magnitude than the original MSS estimate of ≈ -2.4 but still imprecisely estimated. The qualitative direction of MSS is therefore sensitive to whether secular country-level trends are modelled.

Table 3: Specification and identification diagnostics.

Statistic	Value	p -value
<i>Heteroscedasticity</i>		
Breusch–Pagan	142.52	$< 10^{-16}$
<i>Panel structure</i>		
F -test country FE ($H_0: c_i = 0$)	14.90	$< 10^{-16}$
Hausman FE vs RE	0.04	0.998
<i>Weak instruments</i>		
First-stage F baseline (cluster-robust)	2.89	0.056
First-stage F MSS-trends (cluster Wald)	4.45	0.012
First-stage F MSS-trends (homoscedastic)	7.48	0.0006
Stock–Yogo 10%-maximum-size critical value	19.93	—
<i>Endogeneity</i>		
Wu–Hausman	0.14	0.705
<i>Overidentification</i>		
Sargan	2.40	0.121
<i>Weak-IV-robust and bootstrap inference</i>		
Anderson–Rubin 95% CI for β^{IV}	$[-4.41, 7.49]$	—
Cluster-bootstrap 95% CI for β^{IV}	$[-3.52, 3.51]$	—

The diagnostics tell a coherent story. Heteroscedasticity is decisively present, confirming the necessity of robust standard errors for the LPM. Country fixed effects are jointly significant, but the Hausman test returns a vanishing statistic that we discount: the test’s construction assumes the FE estimator is efficient under H_0 , which fails once standard errors are clustered, and its power is limited in moderately unbalanced panels. We default to FE.

The weak-instrument diagnostics are the crux. The first-stage F on the two excluded rainfall instruments is 2.89 in our baseline, roughly one-seventh of the Stock-Yogo threshold of 19.93. The MSS-trends specification raises it to 4.45 (cluster-robust) or 7.48 (homoscedastic), notably stronger but still below the threshold. Conventional 2SLS inference is therefore unreliable in both specifications, and the Wu-Hausman test fails to reject exogeneity ($p = 0.705$): in this sample we have no positive statistical evidence that OLS is biased, though theory says it must be.

Under genuinely weak instruments only weak-IV-robust inference is valid. The Anderson-Rubin 95% confidence set for β is $[-4.41, 7.49]$, uninformative about either sign or magnitude. A non-parametric cluster bootstrap (500 replications, resampling countries with replacement) yields a closely comparable 95% percentile interval of $[-3.52, 3.51]$. The Sargan test on the two rainfall instruments fails to reject ($p = 0.121$), which we read as weak reassurance only.

Table 4: Panel-internal identification strategies. Column 1 is the Anderson and Hsiao (1981) FD-IV on the static structural equation; columns 2 and 3 are two-step difference- and system-GMM on the dynamic specification (3), with collapsed instrument matrices at lags 2–3. Cluster-robust SE at country level in parentheses.

	Anderson–Hsiao	Arellano–Bond	Blundell–Bond
Conflict (lag)		0.334** (0.142)	0.628*** (0.095)
GDP growth	−0.349 (0.383)	−0.293* (0.150)	−0.296 (0.225)
GDP growth (lag)		0.044 (0.211)	−0.046 (0.225)
Polity 2 (lag)	−0.005 (0.005)	0.000 (0.004)	−0.006* (0.003)
N (country-years)	620	—	—
First-stage F on $g_{i,t-1}^y$	451.7	—	—
Sargan–Hansen p	—	0.51	0.09
m_2 (AR(2)) p	—	0.50	0.47

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The three estimators tell a consistent qualitative story under identifying assumptions that differ. Anderson-Hsiao gives $\hat{\beta}_{AH} = -0.349$ with cluster-robust SE 0.383. The first-stage F on the lagged level $g_{i,t-1}^y$ is 451.7, far above the Stock-Yogo 10%-maximum-size critical value of 16.4 for one instrument and one endogenous regressor; but the strength reflects mean reversion in the country-level growth series rather than new identifying variation, since high past growth predicts a large negative change in current growth almost mechanically. Identification rests on $\text{Cov}(g_{i,t-1}^y, \Delta u_{it}) = 0$, which is logically independent from the rainfall exclusion.

The Arellano-Bond difference-GMM estimator, which adds $C_{i,t-1}$ to the equation and uses deeper lags as instruments, returns $\hat{\beta}_{AB} = -0.293$ ($p \approx 0.05$). Blundell-Bond yields a similar point estimate -0.296 but the standard error widens and significance drops below the 10% level. The lagged conflict coefficient is large and highly significant in system-GMM ($\hat{\rho} = 0.628$, $p < 0.001$), confirming substantial persistence and indicating that the static specifications of Table 2 are mis-specified. The Sargan-Hansen tests do not reject for either estimator ($p = 0.51$ for AB; $p = 0.09$ for BB, borderline and read with caution); the m_2 tests do not detect second-order autocorrelation in differences ($p = 0.50$ and 0.47). Both validations support the identifying restrictions.

Across all three panel-internal designs the MSS sign reappears, though the magnitude is pinned down only loosely. None of these strategies uses rainfall.

7 Discussion

The empirical findings are mixed and depend on identification assumptions.

OLS recovers the qualitative MSS direction with a tight standard error: $\hat{\beta}_{\text{POLS}} = -0.507$. Under exogeneity of growth, this would be the most precise summary of the conditional relationship between growth and conflict. As a causal estimate, it is not defensible: the three endogeneity sources of Section 1 rule out the required orthogonality.

Under the MSS rainfall identification, conventional inference is unreliable. The first-stage F of 2.89 in our two-way FE specification is roughly one-seventh of the Stock-Yogo threshold; the 2SLS point estimate carries the wrong sign and the Anderson-Rubin 95% confidence set $[-4.41, 7.49]$ is uninformative about either sign or magnitude. Restoring the MSS country-trend specification raises the first stage to 4.45 and recovers $\hat{\beta} = -1.30$, of the correct sign but still imprecisely estimated. The cluster bootstrap interval $[-3.52, 3.51]$ confirms the order of magnitude of the irreducible uncertainty under conventional 2SLS, though it does not itself restore identification under weak instruments.

Under the panel-internal identification strategies of Sections 5.5 and 5.6, the negative sign reappears across all three estimators ($\hat{\beta}_{\text{AH}} = -0.349$, $\hat{\beta}_{\text{AB}} = -0.293$, $\hat{\beta}_{\text{BB}} = -0.296$), with Arellano-Bond significant at the 10% level. The Anderson-Hsiao first-stage strength is mechanical, driven by mean reversion in growth, but the GMM estimators rest on deeper lag-orthogonality and (for system-GMM) initial-condition restrictions that are logically distinct from the rainfall exclusion.

The data therefore support the qualitative sign of the MSS effect across at least two distinct identifying assumptions (rainfall exclusion, past-growth orthogonality), but do not support a precise magnitude or a credible defence of either assumption against the Sarsons (2015) critique, which targets the rainfall channel directly. The Sargan over-identification test on the two rainfall instruments does not reject, but with one degree of freedom and two instruments that share their exogeneity argument it has neither the power nor the structure to detect the kind of violation that matters. A precise causal estimate of how much income shocks raise the probability of civil conflict in this sample requires a stronger instrument: the commodity-price approach of Brückner and Ciccone (2010) being the leading candidate, or a research design that does not rely on rainfall at all.

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